

PRACTICAL DBMS

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COURSE CODE: CSA0503

COURSE NAME: DATABASE MANAGEMENT SYSTEMS

EXPERIMENT NO:2 DDL Commands with Constraints – PRIMARY, FOREIGN KEY, UNIQUE, CHECK

AIM:

To add the constraints like primary key, foreign key, unique key and check using DDL commands.

Description:

PRIMARY KEY:

The PRIMARY KEY constraint uniquely identifies each record in a database table.

Primary keys must contain UNIQUE values, and cannot contain NULL values.

A table can have only one primary key, which may consist of single or multiple fields.

FOREIGN KEY:

➤ A FOREIGN KEY is a key used to link two tables together.

➤ A FOREIGN KEY is a field (or collection of fields) in one table that refers to the PRIMARY KEY in another table.

➤ The table containing the foreign key is called the child table, and the table containing the candidate key is

called the referenced or parent table.

UNIQUE Constraint:

➤ The UNIQUE constraint ensures that all values in a column are different.

➤ Both the UNIQUE and PRIMARY KEY constraints provide a guarantee for uniqueness for a column

or set of columns.

➤ A PRIMARY KEY constraint automatically has a UNIQUE constraint.

➤ However, you can have many UNIQUE constraints per table, but only one PRIMARY KEY

constraint per table.

CHECK Constraint:

➤ The CHECK constraint is used to limit the value range that can be placed in a column

➤ If you define a CHECK constraint on a single column it allows only certain values for this column.

➤ If you define a CHECK constraint on a table it can limit the values in certain columns based on

values in other columns in the row.

PRIMARY:

ALTER TABLE table_name

ADD PRIMARY KEY(primary_key_column);

```
mysql> alter table student add primary key(regno);
```

```
Query OK, 0 rows affected (0.18 sec)
```

```
Records: 0 Duplicates: 0 Warnings: 0
```

```
mysql> desc student;
```

Field	Type	Null	Key	Default	Extra
regno	int	NO	PRI	NULL	
gender	varchar(15)	YES		NULL	
name	varchar(15)	YES		NULL	

3 rows in set (0.00 sec)

FOREIGN KEY:

ALTER TABLE table_name

ADD CONSTRAINT constraint_name

FOREIGN KEY foreign_key_name (columns)

REFERENCES parent_table(columns)

ON DELETE action

ON UPDATE action

UNIQUE:

```

CREATE TABLE table_1(
...
column_name_1 data_type,
...
UNIQUE(column_name_1)
);

CHECK

CREATE TABLE IF NOT EXISTS parts (
part_no VARCHAR(18) PRIMARY KEY,
description VARCHAR(40),
cost DECIMAL(10 , 2 ) NOT NULL CHECK(cost > 0), price DECIMAL (10,2) NOT
NULL
);

```

```

mysql> alter table student add unique key(name);
Query OK, 0 rows affected (0.15 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> desc student;
+-----+-----+-----+-----+-----+-----+
| Field | Type          | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| regno | int           | NO   | PRI | NULL    |       |
| gender | varchar(15)   | YES  |     | NULL    |       |
| name   | varchar(15)   | YES  | UNI | NULL    |       |
+-----+-----+-----+-----+-----+-----+
3 rows in set (0.01 sec)

```

Result:

DDL Commands with Primary, Foreign, Unique, Check constraints are updated and verified.

